connection between black holes and elementary particles. The road to this connection is a bit circuitous, but it takes us through some of the most interesting developments in string theory, making it a journey well worth taking.

It begins with a seemingly unrelated question that string theorists have kicked around since the late 1980s. Mathematicians and physicists have long known that when six spatial dimensions are curled up into a CalabiYau shape, there are generally two kinds of spheres that are embedded within the shape's fabric. One kind are the two-dimensional spheres, like the surface of a beach ball, that played a vital role in the space-tearing flop transitions of Chapter 11. The other kind are harder to picture but they are equally prevalent. They are three-dimensional spheres—like the surfaces of beach balls adorning the sandy ocean shores of a universe with four extended space dimensions. Of course, as we discussed in Chapter 11, an ordinary beach ball in our world is itself a three-dimensional object, but its surface, just like the surface of a garden hose, is two-dimensional: You need only two numbers—latitude and longitude, for instance—to locate any position on its surface. But we are now imagining having one more space dimension: a four-dimensional beach ball whose surface is three-dimensional. As it's pretty close to impossible to picture such a beach ball in your mind's eye, for the most part we will appeal to lower-dimensional analogs that are more easily visualized. But, as we shall now see, one aspect of the three-dimensional nature of the spherical surfaces is of prime importance.

By studying the equations of string theory, physicists realized that it is possible, and even likely, that as time evolves, these three-dimensional spheres will shrink—collapse—to vanishingly small volume. But what would happen, string theorists asked, if the fabric of space were to collapse in this manner? Will there be some catastrophic effect from this kind of pinching of the spatial fabric? This is much like the question we posed and resolved in Chapter 11, but here we are focusing on collapsing three-dimensional spheres, whereas in Chapter 11 we focused solely on collapsing two-dimensional spheres. (As in Chapter 11, since we are envisioning that a piece of a Calabi-Yau shape is shrinking, as opposed to the whole Calabi-Yau shape itself, the small radius/large radius identification of Chapter 10 does not